

Strategies for Mitigating Antibody-Drug Conjugate Related Adverse Events for Precision Therapy.

PMID: 36383913 | Indexed in PubMed

Antibody-drug conjugates (ADCs) have been revolutionary in improving personalized therapy of cancer. Through combining monoclonal antibodies, which are targeted to tumor-specific antigens, and cytotoxic agents, ADCs lead to selective delivery of active components, also called payloads, to cancerous cells while sparing healthy body cells from possible collateral damage. Adverse events, however, can still develop because of early release of the payload or cross-expression of targets by normal cells leading to collateral damage. In this review, we elaborate on the common and serious adverse events for the currently US Food and Drug Administration-approved ADCs in solid tumors and discuss mitigating strategies and dose modification guidance for optimizing efficacy and toxicity.

Novel formats of antibody conjugates: recent advances in payload diversity, conjugation, and linker chemistry.

PMID: 37953519 | Indexed in PubMed

In the field of bioconjugates, the focus on antibody - drug conjugates (ADCs) with novel payloads beyond the traditional categories of potent cytotoxic agents is increasing. These innovative ADCs exhibit various molecular formats, ranging from small-molecule payloads, such as immune agonists and proteolytic agents, to macromolecular payloads, such as oligonucleotides and proteins. This review offers an in-depth exploration of unconventional strategies for designing conjugates with novel mechanisms of action and notable examples of approaches that show promising prospects. Representative examples of novel format payloads and their classification, attributes, and appropriate conjugation techniques are discussed in detail. The existing basic technologies used to manufacture ADCs can be directly applied to synthesize novel formatted conjugates. However, a wide variety of new payloads require the creation of customized technologies adapted to the unique characteristics of these payloads. Consequently, fundamental technologies, such as conjugation methods aimed at achieving high drug - antibody ratios and developing stable crosslinkers, are likely to become increasingly important research areas in the future.

Pharmacokinetics and pharmacodynamics of antibody-drug conjugates for the treatment of patients with breast cancer.

PMID: 38214896 | Indexed in PubMed

Currently three antibody-drug-conjugates (ADC) are approved by the European Medicines Agency (EMA) for treatment of breast cancer (BC) patient: trastuzumab-emtansine, trastuzumab-deruxtecan and sacituzumab-govitecan. ADC are composed of a monoclonal antibody (mAb) targeting a specific antigen, a cytotoxic payload and a linker. Pharmacokinetics (PK) and pharmacodynamics (PD) distinguish ADC from conventional chemotherapy and must be understood by clinicians. Our review delineates the PK/PD profiles of ADC approved for the treatment of BC with insight for future development. This is an expert opinion literature review based on the EMA's Assessment Reports, enriched by a comprehensive literature search performed on Medline in August 2023. All three ADC distributions are described by a

two-compartment structure: tissue and serum. Payload concentration peak is immediate but remains at low concentration. The distribution varied for all ADC only with body weight. mAb will be metabolised firstly by the saturable complex formation of ADC/Tumour-Receptor and secondly by binding of FcγRs in immune cells. They are all excreted in the bile and faeces with minimal urine elimination. Dose adjustments, apart from weight, are not recommended. Novel ADC are composed of cleavable linkers with various targets/payloads with the same PK/PD properties, but novel structures of ADC are in development.

Natural-source payloads used in the conjugated drugs architecture for cancer therapy: Recent advances and future directions.

PMID: 39134188 | Indexed in PubMed

Drug conjugates are obtained from tumor-located vectors connected to cytotoxic agents via linkers, which are designed to deliver hyper-toxic payloads directly to targeted cancer cells. These drug conjugates include antibody-drug conjugates (ADCs), peptide-drug conjugates (PDCs), small molecule-drug conjugates (SMDCs), nucleic acid aptamer-drug conjugates (ApDCs), and virus-like drug conjugate (VDCs), which show great therapeutic value in the clinic. Drug conjugates consist of a targeting carrier, a linker, and a payload. Payloads are key therapy components. Cytotoxic molecules and their derivatives derived from natural products are commonly used in the payload portion of conjugates. The ideal payload should have sufficient toxicity, stability, coupling sites, and the ability to be released under specific conditions to kill tumor cells. Microtubule protein inhibitors, DNA damage agents, and RNA inhibitors are common cytotoxic molecules. Among these conjugates, cytotoxic molecules of natural origin are summarized based on their mechanism of action, conformational relationships, and the discovery of new derivatives. This paper also mentions some cytotoxic molecules that have the potential to be payloads. It also summarizes the latest technologies and novel conjugates developed in recent years to overcome the shortcomings of ADCs, PDCs, SMDCs, ApDCs, and VDCs. In addition, this paper summarizes the clinical trials conducted on conjugates of these cytotoxic molecules over the last five years. It provides a reference for designing and developing safer and more efficient conjugates.

Beyond cytotoxic potency: disposition features required to design ADC payload.

PMID: 39017706 | Indexed in PubMed

1. Antibody-drug conjugates (ADCs) have demonstrated impressive clinical usefulness in treating several types of cancer, with the notion of widening of the therapeutic index of the cytotoxic payload through the minimisation of the systemic toxicity. Therefore, choosing the most appropriate payload molecule is a particularly important part of the early design phase of ADC development, especially given the highly competitive environment ADCs find themselves in today. 2. The focus of the current review is to describe critical attributes/considerations needed in the discovery and ultimately development of cytotoxic payloads in support of ADC design. In addition to potency, several key dispositional characteristics including solubility, permeability and bystander effect, pharmacokinetics, metabolism, and drug-drug interactions, are described as being an integral part of the integrated activities required in the design of clinically safe and useful ADC therapeutic agents.

Dual-payload antibody-drug conjugates: Taking a dual shot.

PMID: 39481229 | Indexed in PubMed

Antibody-drug conjugates (ADCs) enable the precise delivery of cytotoxic agents by conjugating small-molecule drugs with monoclonal antibodies (mAbs). Over recent decades, ADCs have demonstrated substantial clinical efficacy. However, conventional ADCs often encounter various clinical challenges, including suboptimal efficacy, significant adverse effects, and the development of drug resistance, limiting their broader clinical application. Encouragingly, a next-generation approach-dual-payload ADCs-has emerged as a pioneering strategy to address these challenges. Dual-payload ADCs are characterized by the incorporation of two distinct therapeutic payloads on the same antibody, enhancing treatment efficacy by promoting synergistic effects and reducing the risk of drug resistance. However, the synthesis of dual-payload ADCs is complex due to the presence of multiple functional groups on antibodies. In this review, we comprehensively summarize the construction strategies for dual-payload ADCs, ranging from the design of ADC components to orthogonal chemistry. The subsequent sections explore current challenges and propose prospective strategies, highlighting recent advancements in dual-payload ADC research, thereby laying the foundation for the development of next-generation ADCs.

Antibody drug conjugates beyond cytotoxic payloads.

PMID: 37981349 | Indexed in PubMed

For many years, antibody drug conjugates (ADC) have teased with the promise of targeted payload delivery to diseased cells, embracing the targeting of the antibody to which a cytotoxic payload is conjugated. During the past decade this promise has started to be realised with the approval of more than a dozen ADCs for the treatment of various cancers. Of these ADCs, brentuximab vedotin really laid the foundations of a template for a successful ADC with lysosomal payload release from a cleavable dipeptide linker, measured DAR by conjugation to the Cys-Cys interchain bonds of the antibody and a cytotoxic payload. Using this ADC design model oncology has now expanded their repertoire of payloads to include non-cytotoxic compounds. These new payload classes have their origins in prior medicinal chemistry programmes aiming to design selective oral small molecule drugs. While this may not have been achieved, the resulting compounds provide excellent starting points for ADC programmes with some compounds amenable to immediate linker attachment while for others extensive SAR and structural information offer invaluable design insights. Many of these new oncology payload classes are of interest to other therapeutic areas facilitating rapid access to drug-linkers for exploration as non-oncology ADCs. Other therapeutic areas have also pursued unique payload classes with glucocorticoid receptor modulators (GRM) being the most clinically advanced in immunology. Here, ADC payloads come full circle, as oncology is now investigating GRM payloads for the treatment of cancer. This chapter aims to cover all these new ADC approaches while describing the medicinal chemistry origins of the new non-cytotoxic payloads.

Payload diversification: a key step in the development of antibody-drug conjugates.

PMID: 36650546 | Indexed in PubMed

Antibody-drug conjugates (ADCs) is a fast moving class of targeted biotherapeutics that currently

combines the selectivity of monoclonal antibodies with the potency of a payload consisting of cytotoxic agents. For many years microtubule targeting and DNA-intercalating agents were at the forefront of ADC development. The recent approval and clinical success of trastuzumab deruxtecan (Enhertu®) and sacituzumab govitecan (Trodelvy®), two topoisomerase 1 inhibitor-based ADCs, has shown the potential of conjugating unconventional payloads with differentiated mechanisms of action. Among future developments in the ADC field, payload diversification is expected to play a key role as illustrated by a growing number of preclinical and clinical stage unconventional payload-conjugated ADCs. This review presents a comprehensive overview of validated, forgotten and newly developed payloads with different mechanisms of action.

Natural products as exquisitely potent cytotoxic payloads for antibody- drug conjugates.

PMID: 25487009 | Indexed in PubMed

Antibody-drug conjugates (ADCs) are an emerging area of study within medicinal chemistry and are thought of as sophisticated drug delivery systems due to their specificity to a disease-targeted antigen. ADCs have been actively utilized as therapeutics for hematological and solid tumor cancers due to their capability to deliver a cytotoxic compound to a specific cancer cell without affecting normal cells. An antibody drug conjugate has three major constituents: a monoclonal antibody (mAb), a chemical linker, and a potent cytotoxic payload. There has been a continuing effort to optimize antibody-drug conjugates, with the primary focus of design and development directed at either the mAb or the chemical linker, with little effort devoted to the optimization of payload compounds. In fact, among the 114 ongoing or recently completed clinical trials, there is generally a lack of diversity in the cytotoxic payloads that are utilized, with only seven payload compounds reported (four additional trials are ongoing with structures that have not been reported). Six of these seven payload compounds are derived from natural product sources, highlighting the importance of natural products as cytotoxic payloads for ADC.

A Systems Pharmacology Model for Drug Delivery to Solid Tumors by Antibody-Drug Conjugates: Implications for Bystander Effects.

PMID: 31828446 | Indexed in PubMed

Antibody-drug conjugates (ADCs) are cancer drugs composed of a humanized antibody linked to a cytotoxic payload, allowing preferential release of payload in cancer cells expressing the antibody-targeted antigen. Here, a systems pharmacology model is used to simulate ADC transport from blood to tumor tissue and ADC uptake by tumor cells. The model includes effects of spatial gradients in drug concentration in a three-dimensional network of tumor blood vessels with realistic geometry and accounts for diffusion of ADC in the tumor extracellular space, binding to antigen, internalization, intracellular processing, and payload efflux from cells. Cells that process an internalized ADC-antigen complex may release payload that can be taken up by other "bystander" cells. Such bystander effects are included in the model. The model is used to simulate conditions in previous experiments, showing good agreement with experimental results. Simulations are used to analyze the relationship of bystander effects to payload properties and single-dose administrations. The model indicates that exposure of payload to cells distant from vessels is sensitive to the free payload diffusivity in the extracellular space.

When antigen expression is heterogeneous, the model indicates that the amount of payload accumulating in non-antigen-expressing cells increases linearly with dose but depends only weakly on the percentage of antigen-expressing cells. The model provides an integrated mechanistic framework for understanding the effects of spatial gradients on drug distribution using ADCs and for designing ADCs to achieve more effective payload distribution in solid tumors, thereby increasing the therapeutic index of the ADC.
